

lecture 6

DTFT analysis equation $\Rightarrow X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$

- synthesis equation $\Rightarrow$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} \cdot d\omega$$

DTFT of basic signals:

$$x[n] \xrightarrow{\text{DTFT}} X(e^{j\omega})$$

$$\delta[n] \rightarrow 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$$

$$u[n] \rightarrow \frac{1}{1 + e^{-j\omega}} + \pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$$

$$\text{sgn}[n] \rightarrow \frac{2}{1 + e^{-j\omega}}$$

$$a^n u[n] \rightarrow \frac{1}{1 - a e^{-j\omega}}, \quad |a| < 1$$

$$-a^n u[-n-1] \rightarrow \frac{1}{1 - a e^{-j\omega}}, \quad |a| > 1$$

$$\sin(\omega_0 n) \rightarrow \frac{\pi}{j} \sum_k \delta(\omega - \omega_0 - 2\pi k) + \delta(\omega + \omega_0 - 2\pi k)$$

$$\cos(\omega_0 n) \rightarrow \pi \sum_k [\delta(\omega - \omega_0 - 2\pi k) + \delta(\omega + \omega_0 - 2\pi k)]$$

DTFT Properties:-

Property	$x[n]$	$X(e^{j\omega})$
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Periodic $\rightarrow x[n] \rightarrow X(e^{j(\omega+2\pi)}) = X(e^{j\omega})$

Linear $\rightarrow \alpha_1 x_1[n] + \alpha_2 x_2[n] \rightarrow \alpha_1 X_1(e^{j\omega}) + \alpha_2 X_2(e^{j\omega})$

Time Shift $\rightarrow x[n-k] \rightarrow e^{-j\omega k} X(e^{j\omega})$

Convolution $\rightarrow x[n] * h[n] \rightarrow X(e^{j\omega}) H(e^{j\omega})$

Modulation $\rightarrow x[n] e^{j\omega_0 n} \rightarrow X(e^{j(\omega - \omega_0)})$

Parseval $\rightarrow \sum_{n=-\infty}^{\infty} |x[n]|^2 \rightarrow \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$



Inverse DTFT (Synthesis equation)

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

usually we don't use this rule we use properties
any basic signal DTFT to get the

inverse in many cases we use partial fraction.

Example: find the inverse DTFT of the signal

$$X(e^{j\omega}) = \frac{3}{1 - 0.2e^{j\omega}} + \frac{5}{1 - 4e^{-j\omega}}$$

by using linearity property:-

$$x[n] = 3(0.2)^n u[n] - 5(4)^n u[-n-1]$$

Generic Rule $\Rightarrow \frac{x}{1 - y e^{j\omega}} \Rightarrow x \cdot (y)^n u[n]$

Example: $X(e^{j\omega}) = \frac{3e^{-j2\omega} + 5e^{-j3\omega}}{1 + 0.3e^{-j\omega}}$

$$X(e^{j\omega}) = \frac{3e^{-j2\omega}}{1 + 0.3e^{-j\omega}} + \frac{5e^{-j3\omega}}{1 + 0.3e^{-j\omega}}$$

$$= 3(-0.3)^{n-2} u[n-2] + 5(-0.3)^{n-3} u[n-3]$$

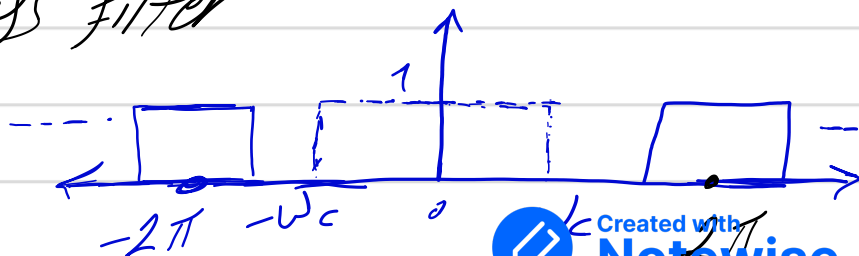
* So we got this solution from time shift and linear property.



Filters:- Filter is a system that passes some frequency components and stop other frequencies components.

Ideal filter $\Rightarrow$ Is a filter that multiplies the pass-band component by 1 and stop band component by 0.

Example: Ideal Low Pass filter



in frequency domain

$$\text{Frequency response} = H(e^{j\omega}) = \begin{cases} 1, & |\omega| < \omega_c \\ 0, & \text{o.w} \end{cases}$$

* In Time Domain :- $h[n] = \text{DTF}^{-1} \{ H(e^{j\omega}) \} =$
 $\frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\omega}) e^{j\omega n} \cdot d\omega$

$$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} (1) e^{j\omega n} \cdot d\omega = \frac{1}{2\pi} \left[\frac{e^{j\omega n}}{jn} \right]_{-\omega_c}^{\omega_c}$$

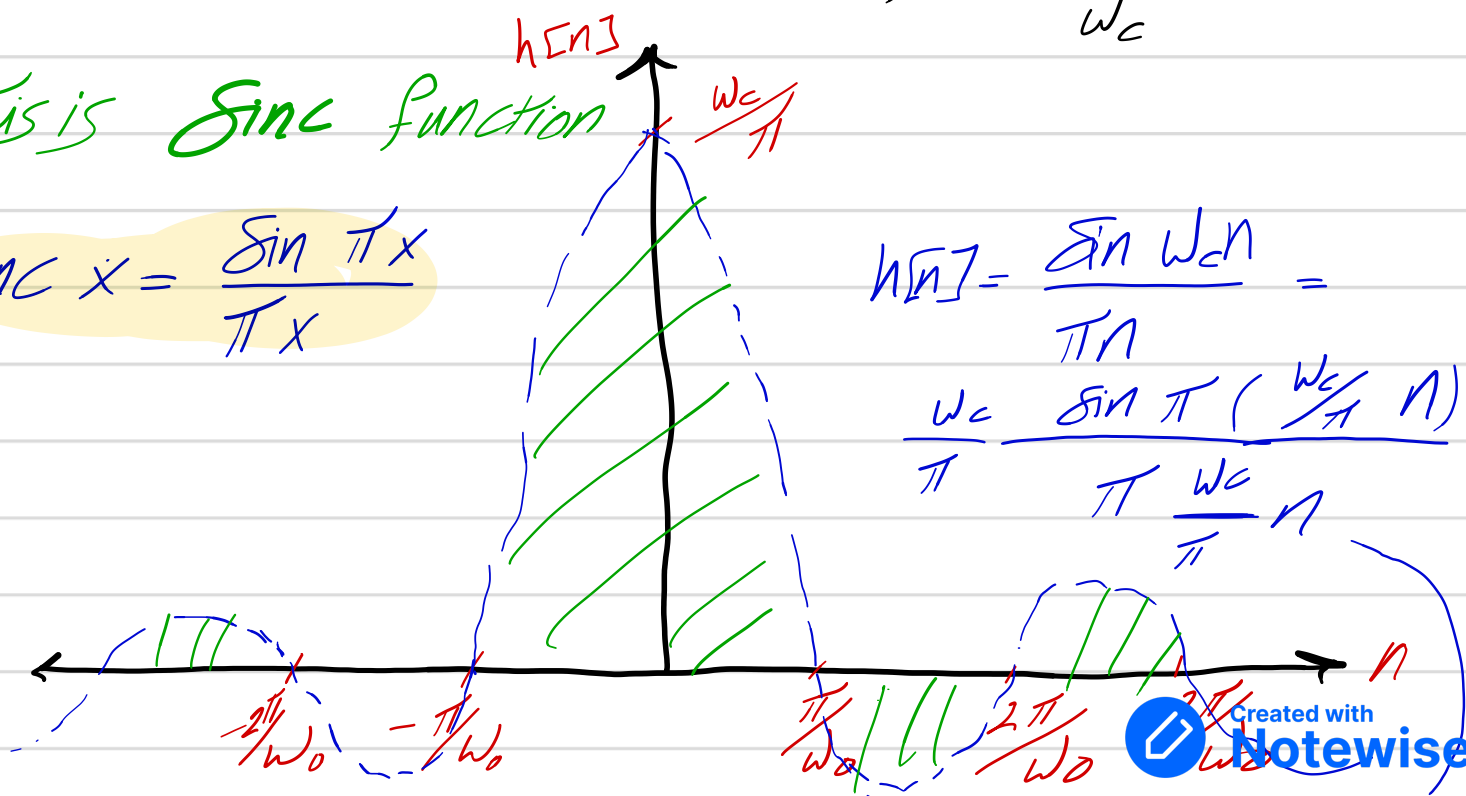
$$= \frac{2 \sin \omega_c n}{2\pi jn} = \frac{\sin \omega_c n}{\pi n}$$

$\sin \omega_c n = 0$ when $\omega_c n = K\pi$, $n = \frac{K\pi}{\omega_c}$

This is Sinc function

$$\text{Sinc } x = \frac{\sin \pi x}{\pi x}$$

$$h[n] = \frac{\sin \omega_c n}{\pi n} = \frac{\omega_c}{\pi} \frac{\sin \pi \left(\frac{\omega_c}{\pi} n \right)}{\pi \frac{\omega_c}{\pi} n}$$



$$h[n] = \frac{w_c}{\pi} \text{sinc}\left(\frac{w_c}{\pi} n\right)$$

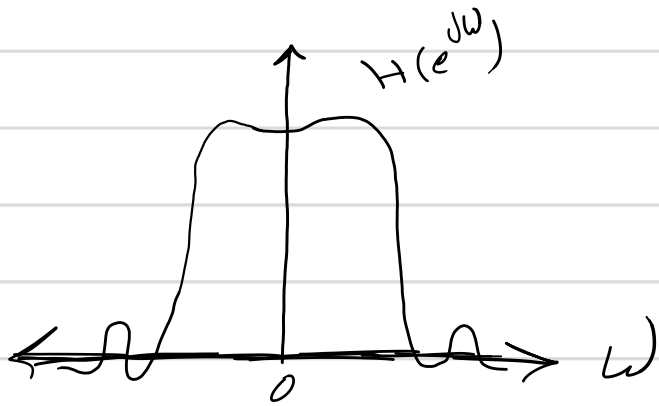
this filter is Ideal because it's anti-causal with infinite support

also this filter has zero phase.

Approximate low pass filters:-

Ideal: Truncation of $h[n]$

e.g. $\Rightarrow h[n] = \begin{cases} h[n] & -N \leq n \leq N \\ 0 & \text{o.w.} \end{cases}$



$$= h[n] \underset{\substack{\uparrow \\ \text{Window function}}}{w[n]}$$

where $w[n] = \begin{cases} 1 & -N \leq n \leq N \\ 0 & \text{o.w.} \end{cases}$

$$\tilde{H}(e^{j\omega}) = H(e^{j\omega}) * w(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\sigma}) e^{j(\omega-\sigma)} d\sigma$$

$$W[n] = \begin{cases} 1 & -N \leq n \leq N \\ 0 & \text{o.w.} \end{cases}$$

$$W(e^{j\omega}) = \sum_{n=-\infty}^{\infty} W[n] e^{-j\omega n} = \sum_{n=-N}^N (1) e^{-j\omega n}$$

$$= e^{j\omega N} + e^{j\omega(N-1)} + \dots + e^{-j\omega N}$$

$$= e^{j\omega N} \left[1 + e^{-j\omega} + \dots + e^{-j2\omega N} \right]$$

$$= e^{j\omega N} \frac{1 - e^{-j\omega(2N+1)}}{1 - e^{-j\omega}}$$

$$= e^{j\omega N} \frac{e^{-j(N+1/2)\omega} (e^{j(N+1/2)\omega} - e^{-j(N+1/2)\omega})}{e^{-j\omega/2} (e^{j\omega/2} - e^{-j\omega/2})}$$

$$e^{-j\omega/2} [e^{j\omega/2} - e^{-j\omega/2}]$$

$$\cancel{e^{-j\omega/2}} \cdot \cancel{2j} \sin[(N+1/2)\omega]$$

$$= \frac{\cancel{e^{-j\omega/2}} \cdot \cancel{2j} \sin \omega/2}{\cancel{e^{-j\omega/2}} \cdot \cancel{2j} \sin \omega/2}$$

$$= \frac{\sin[(N+1/2)\omega]}{\sin \omega/2}$$

$$\text{When } \omega=0 \Rightarrow W(e^{j\omega}) = \frac{N+1/2}{1/2} = 2N+1$$